

1. BEMAC Class B (non-corner)

Binary Erasure MAC

$Z = X + Y$, $Z \in \{0, 1, 2\}$ (deterministic)

Path: $\text{make_path}(N, N/2) = [0]^{N/2} [1]^N [0]^{N/2}$

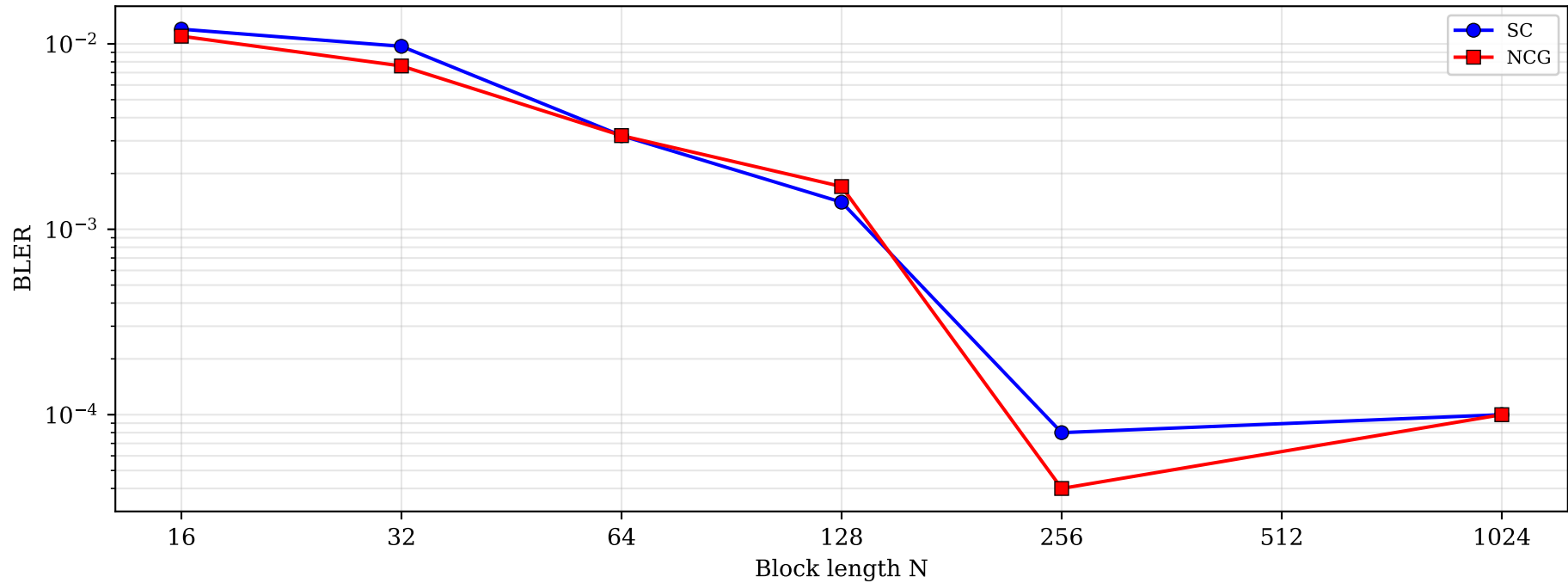
Capacity: $I(X;Z)=0.500$, $I(Y;Z|X)=1.000$, Symmetric point: (0.750, 0.750)

Operating rate: $R_U \sim 0.50$, $R_V \sim 0.70$

NCG: $d=16$, $\text{hidden}=64$, BEMAC vocab=3 embedding

N	ku	kv	SC	SC errs/CW	NCG	NCG errs/CW	NCG/SC
16	8	11	0.0122	122/10K	0.0108	108/10K	0.89x
32	16	22	0.0097	146/15K	0.0076	114/15K	0.78x
64	32	45	0.0032	128/40K	0.0032	154/48K	1.00x
128	64	90	0.0014	129/90K	0.0017	86/50K	1.21x
256	128	179	8e-5	~4/50K	4e-5	~2/50K	0.50x
512	256	358	0.0	0/2K	0.0	0/2K	--
1024	512	716	1e-4	~1/10K	1e-4	~1/10K	1.00x

BLER vs N -- NCG matches or beats SC at all N. No wall.



2. BEMAC Class C (corner)

Binary Erasure MAC

$Z = X + Y$, $Z \in \{0, 1, 2\}$ (deterministic)

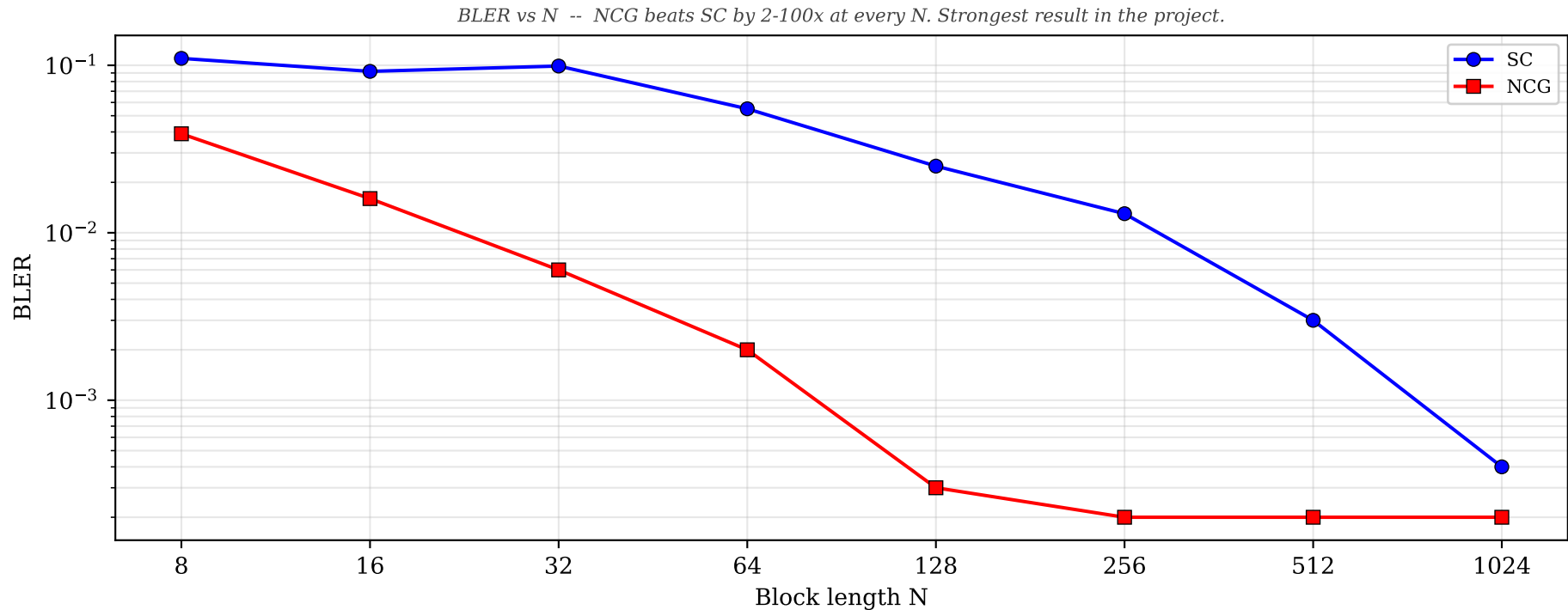
Path: $\text{make_path}(N, N) = [0]^N [1]^N$

Capacity: Corner $(R_U, R_V) = (0.500, 1.000)$

Operating rate: $R_U \sim 0.30$, $R_V \sim 0.60$

NCG: $d=16$, hidden=64, MI-designed frozen set (co-adapted with decoder)

N	ku	kv	SC	SC errs/CW	NCG	NCG errs/CW	NCG/SC
8	2	5	0.1100	329/3K	0.0390	118/3K	0.36x
16	5	10	0.0920	276/3K	0.0160	49/3K	0.18x
32	10	19	0.0990	298/3K	0.0060	17/3K	0.06x
64	19	38	0.0550	165/3K	0.0020	5/3K	0.03x
128	38	77	0.0246	491/20K	3e-4	5/20K	0.01x
256	77	154	0.0134	668/50K	2e-4	9/50K	0.01x
512	154	307	0.0034	168/50K	2e-4	10/50K	0.06x
1024	307	614	4e-4	26/64K	2e-4	5/30K	0.42x



3. GMAC Class B (non-corner)

Gaussian MAC

$$Z = (1-2X) + (1-2Y) + W, \quad W \sim N(0, \sigma^2), \quad \text{SNR} = 6 \text{ dB}$$

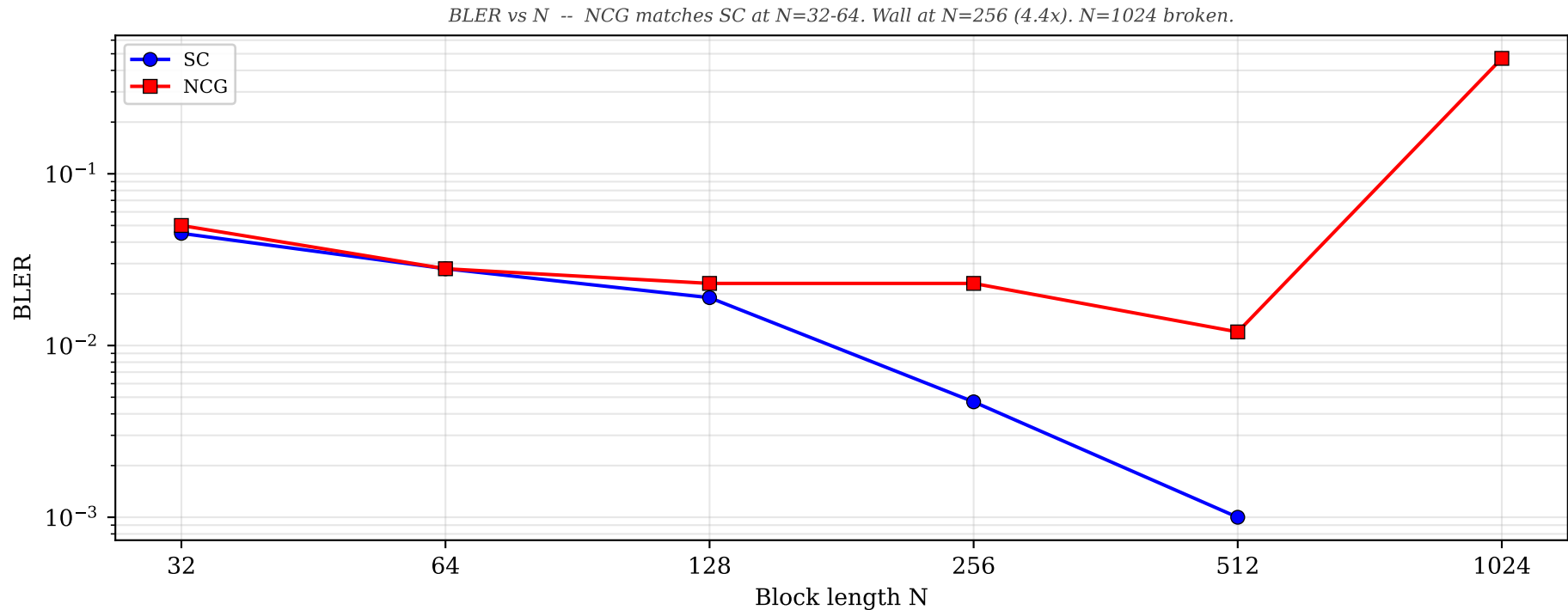
Path: make_path(N, N//2)

Capacity: $I(X;Z)=0.465$, $I(Y;Z|X)=0.912$, $I(X,Y;Z)=1.376$

Operating rate: $R_U = R_V \sim 0.48$ (~70% of symmetric capacity)

NCG: d=16, hidden=64, sequential training

N	ku=kv	SC	SC errs/CW	NCG	NCG errs/CW	NCG/SC
32	15	0.0450	135/3K	0.0503	151/3K	1.12x
64	31	0.0276	138/5K	0.0282	141/5K	1.02x
128	62	0.0187	112/6K	0.0230	46/2K	1.23x
256	123	0.0047	165/35K	0.0230	115/5K	4.9x (wall)
512	246	~0.001	est.	0.0123	123/10K	12x
1024	491	--	--	0.4700	940/2K	BROKEN



4. GMAC Class C (corner)

Gaussian MAC

$$Z = (1-2X) + (1-2Y) + W, \quad W \sim N(0, \sigma^2), \quad \text{SNR} = 6 \text{ dB}$$

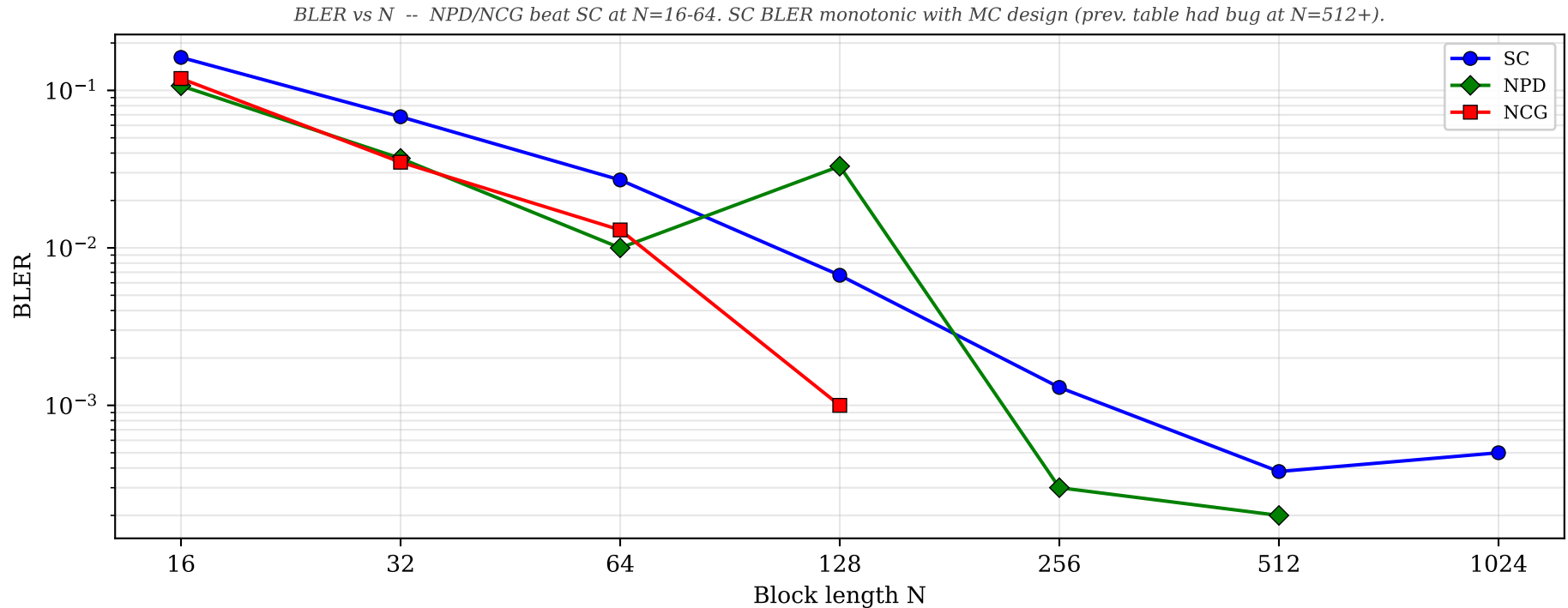
Path: make_path(N, N), corner rate ($R_U \sim 0.23$, $R_V \sim 0.45$)

Capacity: Corner (R_U , R_V) = (0.465, 0.912)

NPD/NCG use MI-designed frozen sets (different from SC design)

SC (MC): uses MC-based frozen set design. SC BLER decreases monotonically.

N	ku	kv	SC(MC)	SC errs/CW	NPD	NPD errs/CW	NCG	NCG errs/CW
16	4	7	0.1620	1620/10K	0.1070	1070/10K	0.1190	~?/1K
32	7	15	0.0681	681/10K	0.0373	373/10K	0.0353	106/3K
64	15	29	0.0273	273/10K	0.0100	100/10K	0.0133	40/3K
128	30	58	0.0067	134/20K	0.0329	329/10K	0.0010	~?/1K
256	59	117	0.0013	126/100K	3e-4	6/20K	--	--
512	119	233	4.0e-4	40/100K	2e-4	11/50K	--	--
1024	238	467	5e-4	10/20K	0.0	0/50K	--	--



5. ABNMAC Class B (non-corner)

Asymmetric Binary Noise MAC

$Z = (X \text{ xor } E_x, Y \text{ xor } E_y)$, correlated binary noise

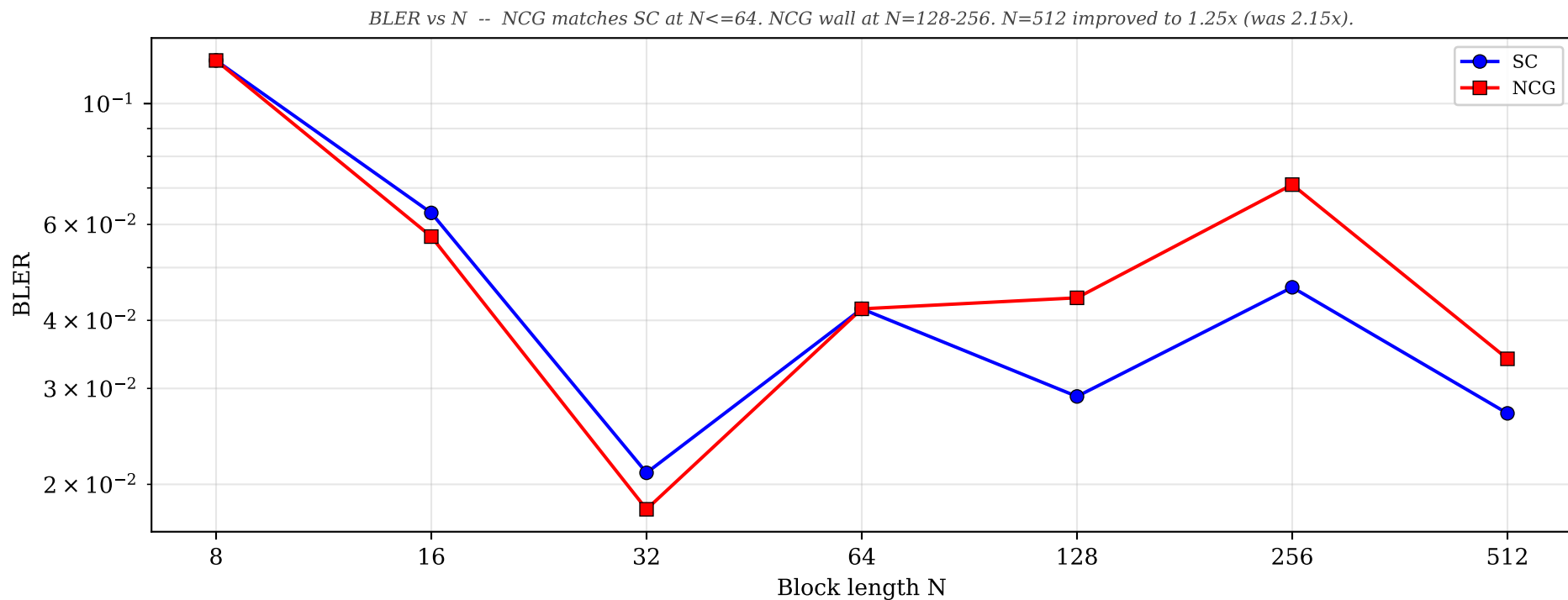
Path: make_path(N, N//2)

Capacity: $I(X;Z) \sim 0.400$, $I(Y;Z|X) \sim 0.800$, Symmetric: (0.600, 0.600)

Operating rate: $R_U \sim R_V \sim 0.30$ ($k_u=k_v$)

NCG: d=16, hidden=64, curriculum from N=8 to N=512

N	ku=kv	SC	SC errs/CW	NCG	NCG errs/CW	NCG/SC
8	3	0.1198	1198/10K	0.1202	1202/10K	1.00x
16	5	0.0629	629/10K	0.0570	570/10K	0.91x
32	10	0.0213	213/10K	0.0182	182/10K	0.85x
64	22	0.0416	416/10K	0.0415	415/10K	1.00x
128	45	0.0288	115/4K	0.0444	222/5K	1.54x
256	102	0.0464	325/7K	0.0713	499/7K	1.54x
512	205	0.0273	191/7K	0.0340	17/500	1.25x



6. ABNMAC Class C (corner)

Asymmetric Binary Noise MAC — SC only

$Z = (X \text{ xor } E_x, Y \text{ xor } E_y)$, correlated binary noise

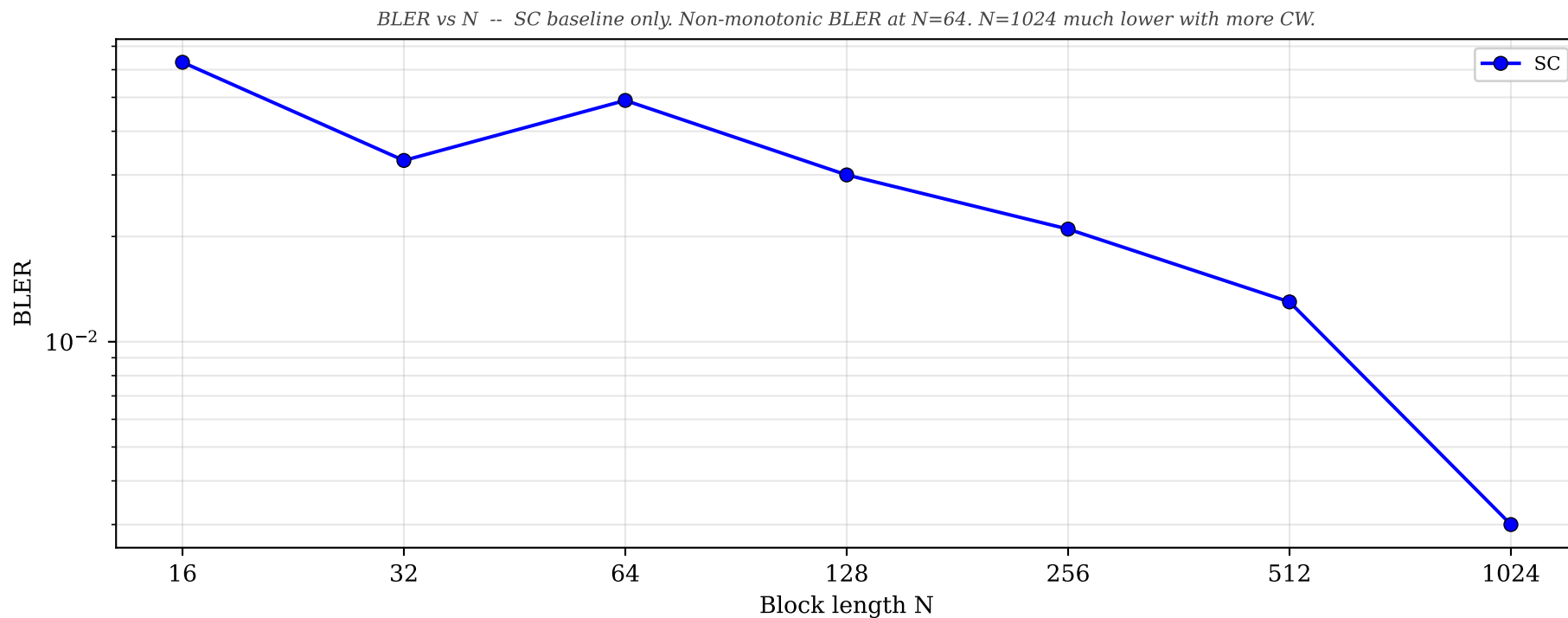
Path: $\text{make_path}(N, N) = [0]^N [1]^N$

Capacity: Corner $(R_U, R_V) = (0.400, 0.800)$

Operating rate: $R_U \sim 0.19, R_V \sim 0.38$

No neural decoder (ABNMAC 2D output incompatible with $z_encoder$)

N	ku	kv	SC	SC errs/CW
16	3	6	0.0626	313/5K
32	6	13	0.0334	167/5K
64	13	26	0.0488	488/10K
128	26	51	0.0300	150/5K
256	51	102	0.0206	103/5K
512	102	205	0.0133	200/15K
1024	205	410	0.0030	59/20K



7. ISI-MAC Class C (corner) — Memory Channel

Inter-Symbol Interference MAC

$$Z[i] = (1-2X[i]) + (1-2Y[i]) + 0.3*(1-2X[i-1]) + 0.3*(1-2Y[i-1]) + W[i]$$

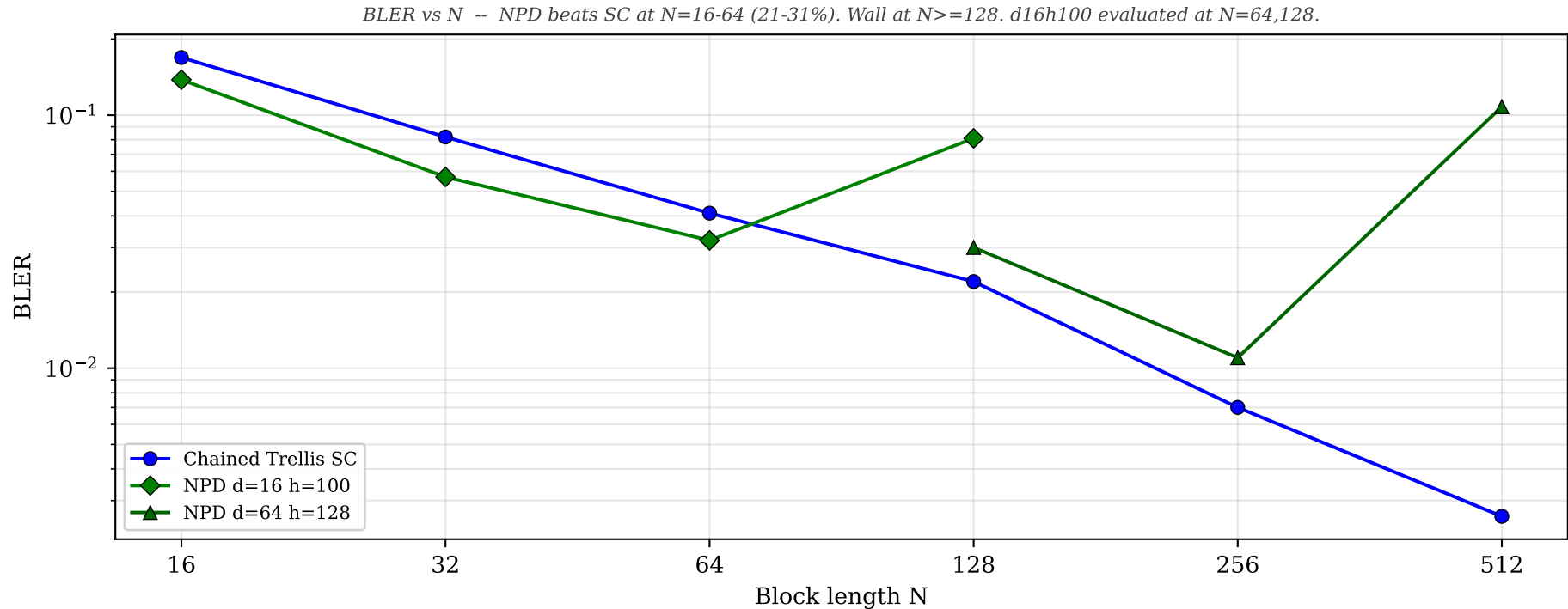
Parameters: $h=0.3$, SNR=6.0 dB ($\sigma^2=0.251$)

Path: make_path(N, N), chained decoding (Stage 1: U, Stage 2: V|U)

Design: GMAC Class C proxy frozen set

SC: chained 2-stage trellis ($|S|=2$ per stage). NPD: chained neural decoder.

N	ku	kv	Chained SC	SC errs/CW	NPD (best)	NPD errs/CW	NPD arch	NPD/SC
16	4	7	0.1689	1689/10K	0.1376	688/5K	d=16 h=100	0.82x
32	7	15	0.0822	822/10K	0.0566	283/5K	d=16 h=100	0.69x
64	15	29	0.0407	407/10K	0.0322	161/5K	d=16 h=100	0.79x
128	30	58	0.0223	223/10K	0.0812	406/5K	d=16 h=100	3.64x
256	59	117	0.0070	105/15K	0.0110	56/5K	d=64 h=128	1.57x
512	119	233	0.0026	51/20K	0.1080	565/5K	d=64 h=128	42x (wall)



8. Ising MAC Class C (corner) — Memory Channel

Ising Channel with Markov State

Good state: $Z=(1-2X)+(1-2Y)+W$. Bad state: $Z=W$. $p_{\text{flip}}=0.1$

Parameters: $\sigma^2=0.251$, $p_{\text{flip}}=0.1$

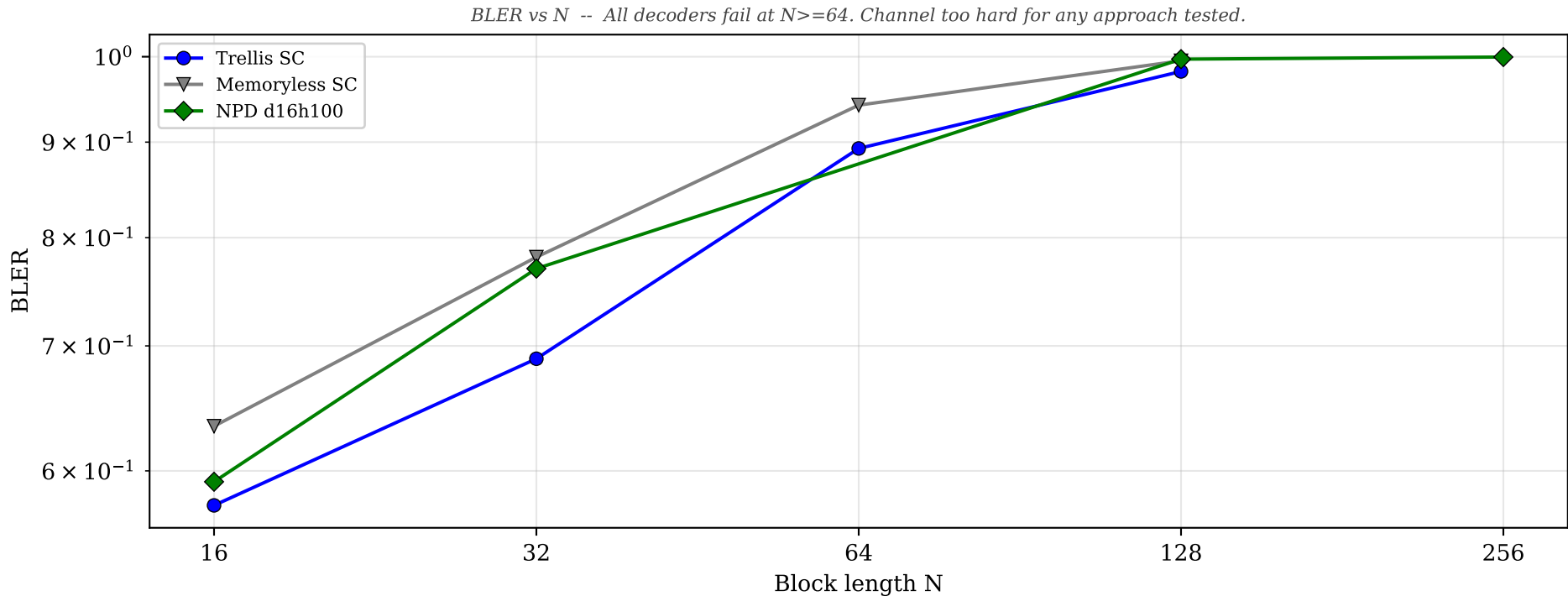
Path: `make_path(N, N)`, chained decoding

Design: GMAC Class C proxy

Extremely hard channel. All decoders fail at $N \geq 64$ (BLER $> 89\%$).

NPD d16h100; trained at $N=16..256$. No better than SC at any N .

N	ku	kv	Trellis SC	SC errs/CW	Memoryless SC	Memless errs/CW	NPD d16h100	NPD errs/CW
16	4	7	0.5750	2873/5K	0.6340	1902/3K	0.5920	2960/5K
32	7	15	0.6890	3443/5K	0.7810	2342/3K	0.7700	3850/5K
64	15	29	0.8930	4465/5K	0.9424	4712/5K	~1.0	--
128	30	58	0.9824	4912/5K	0.9946	4973/5K	0.9970	4985/5K
256	59	117	--	--	--	--	0.9996	4998/5K



8b. Ising MAC Class C — p_flip=0.001 (weaker memory)

Ising Channel with Markov State

Good state: $Z=(1-2X)+(1-2Y)+W$. Bad state: $Z=W$. $p_{\text{flip}}=0.001$

Parameters: $\sigma^2=0.251$, $p_{\text{flip}}=0.001$ (vs 0.1 in 8a)

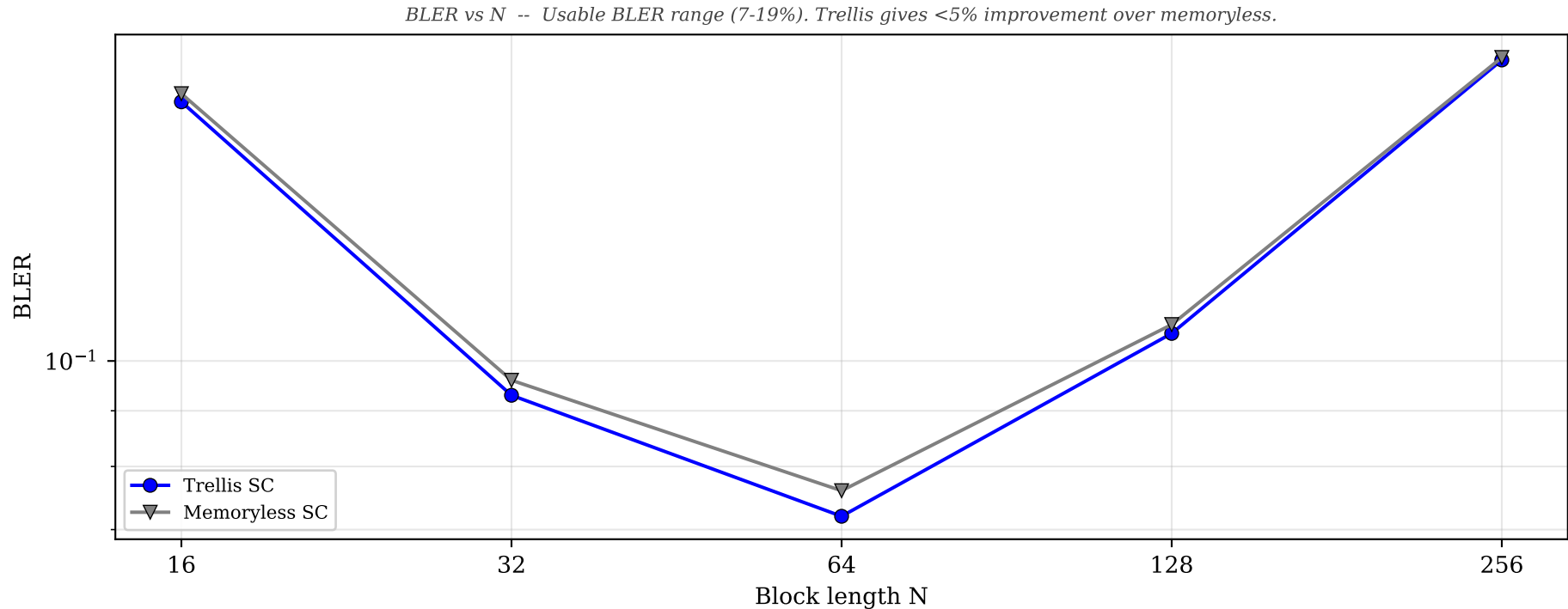
Path: `make_path(N, N)`, chained decoding

Design: GMAC Class C proxy

With rare state flips, Ising $\rightarrow$ near-GMAC. Trellis barely beats memoryless.

Non-monotonic BLER: code rate outpaces coding gain at large N.

N	ku	kv	Trellis SC	SC errs/CW	Memoryless SC	Memless errs/CW
16	4	7	0.1727	1727/10K	0.1756	1756/10K
32	7	15	0.0926	926/10K	0.0958	958/10K
64	15	29	0.0720	720/10K	0.0757	757/10K
128	30	58	0.1056	528/5K	0.1080	540/5K
256	59	117	0.1888	944/5K	0.1896	948/5K



9. MA-AGN MAC Class C (corner) — Continuous-State Memory

Moving-Average Gaussian Noise MAC

$$Z[i] = (1-2X[i]) + (1-2Y[i]) + N[i], \quad N[i] = 0.3*N[i-1] + W[i] \quad (\text{AR}(1) \text{ noise})$$

Parameters: $\alpha=0.3$, $\sigma^2=0.251$

Path: `make_path(N, N)`, chained decoding

Design: GMAC Class C proxy

No analytical trellis (continuous state). Memoryless SC is the baseline.

NPD d16h100 trained for $N=64/128/256$. Beats SC only at $N=16$.

N	ku	kv	Memoryless SC	SC errs/CW	NPD (d16h100)	NPD errs/CW	NPD/SC
16	4	7	0.1750	349/2K	0.1375	275/2K	0.79x
32	7	15	0.0570	114/2K	0.0530	265/5K	0.93x
64	15	29	0.0250	75/3K	0.0354	177/5K	1.42x
128	30	58	0.0066	119/18K	0.1048	524/5K	15.9x
256	59	117	2e-4	1/5K	0.0218	109/5K	109x
512	119	233	8e-4	4/5K	--	--	--

